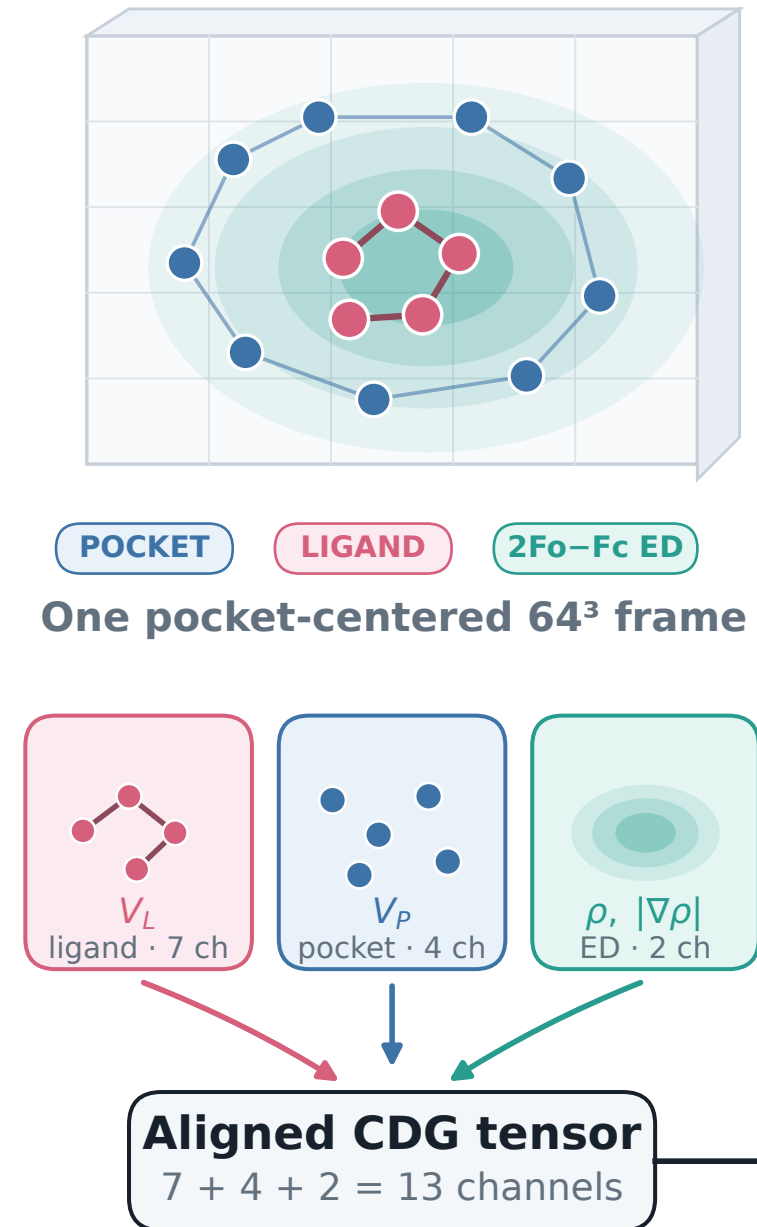


Single-encoder CDG pipeline

Pocket coordinates, ligand coordinates, and experimental electron density are learned jointly in one spatial encoder

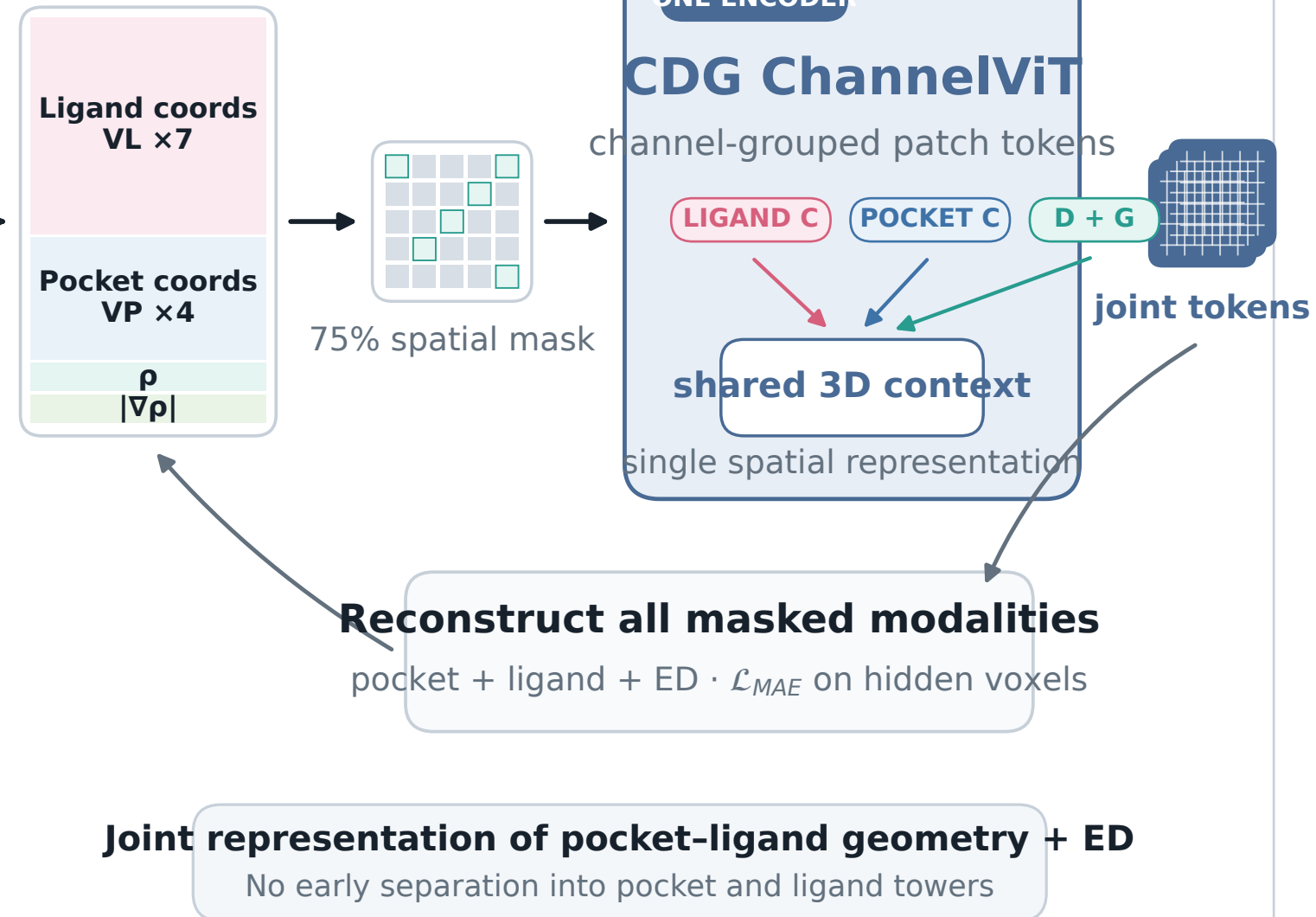
(a) Aligned input data



C: coordinates · D: density · G: density gradient

(b) Joint CDG pre-training

Joint CDG input · 13 ch



(c) Downstream reuse

